



Project Title:

Transjakarta: Jakarta Public Bus Interconnectivity and Coverage

FIT5147 - Data Exploration and Visualization
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Introduction

Jakarta's Transjakarta system is currently the **longest BRT system in the world** (Seasia Stats, 2024), providing vital transport for millions. However, with **Greater Jakarta** expected to become the **world's largest metropolitan area by 2030** (Jauregui-Fung, 2022), ensuring efficient interconnectivity and accessibility remains a major challenge. To keep pace with rising demand, data-driven insights are essential for guiding system improvements.

Motivation

As a frequent Transjakarta user, I've seen firsthand its role in improving urban mobility. This project aims to assess **how well the system meets public needs and identify areas for optimization**. By analyzing route efficiency, user profiles, and infrastructure readiness, I hope to **provide insights that support transport policy and planning** in Jakarta and eventually other growing cities.

Questions

1. How do bus stop usage patterns and traffic volumes affect route efficiency?
2. What are the user profiles across routes, and how can this inform service improvements?
3. Which bus stops require upgrades to accommodate increasing passenger demand?

Data Wrangling and Checking

Data Description and Collection

A. Transjakarta - Public Transportation Transaction, April 2023.

This **Tabular CSV** dataset contains 189,500 rows and 22 columns of transaction data, including user profiles (ID, name, gender, birth year, payment details) and travel details (tap-in/out stops, routes, coordinates, timestamps, and payment amounts). This is a dummy data with user profile and time are generated using Faker. Routes are randomly assigned based on Transjakarta public data.

Since the dataset is only accessible through Python-based APIs such as *kagglehub* and *mlcroissant*, not directly available in *R*, I manually downloaded the *dfTransjakarta180kRows.csv* file from Kaggle and imported it using *read_csv()* for efficiency.

(<https://www.kaggle.com/datasets/dikisahkan/transjakarta-transportation-transaction/>)

B. Transjakarta Greater Jakarta Location Data, updated Jan 2023.

This HTML-based dataset includes 210 rows and 7 columns detailing bus stop locations, corridor names, routes, year built, data period, and remarks.

The data was retrieved from Jakarta's official open-data platform (SatuData Jakarta) using *R*'s *httr* and *jsonlite* packages using methods provided in Anand KC (2019). The **JSON** is **parsed directly from a live API endpoint**, ensuring the dataset remains automatically up to date.

(https://satudata.jakarta.go.id/open-data/detail?kategori=dataset&page_url=data-lokasi-transjakarta-dki-jakarta&data_no=1)

Data Checking and Wrangling

Note: All data processing and analysis were conducted using R Studio.

- Corridor Data

This project focuses on supporting future planning and bus stop improvements. To achieve this, both datasets must be aligned—particularly by matching the Transjakarta Transaction dataset (Dataset 1) with the Transjakarta Location dataset (Dataset 2) using corridor information.

Dataset 1 was filtered to include only major corridors (IDs 1 - 12), as defined in Dataset 2. Since Dataset 2 uses a "Koridor X" format, corridor numbers were extracted using string manipulation (*gsub()* in R) to enable matching. Corridor ID was chosen as the key for merging, as it is central to both datasets.

Initial checks using `distinct()` revealed **inconsistencies errors**:

- Some records had missing `corridorName` values for valid `corridorIDs` (1 - 12).
- Others had missing `corridorID` for known `corridorName` (e.g., "Blok M – Kota").

These issues were treated as **Missing At Random (MAR)**, likely caused by **system or entry errors**. A lookup table was created to cross-reference and correct these mismatches using left joins. After cleaning and filtering, **12,619 records remained**.

- Bus Stop Data

Similar errors were found in tap-in and tap-out stop data. For instance, "Dukuh Atas 2" appeared with two different stop IDs. Using **frequency-based (mode) imputation** (Memon et al., 2023), the most common ID was retained, and others were removed to maintain consistency.

Dataset 1 was then filtered to **only include stops listed in Dataset 2**. This resulted in **1,653 valid records**.

- User Profile Data

User-related fields such as `transID`, `payCardID`, `payCardName`, `payCardSex`, `payCardBirthDate`, and `payCardBank` showed **no duplicates or invalid/missing values**. Each `payCardID` was **uniquely tied to a user**.

- Geolocation Data

Latitude and longitude values for bus stops were complete and showed **no duplication or formatting issues**.

- Transaction Data

No errors were found in tap-in stop combinations. However, several records had **missing** `tapOutStop`, `tapOutStopName`, and `tapOutTime`, despite having a valid `payAmount`.

These were treated as **MAR**—potentially due to **users exiting without tapping out or machine failures**. Since these errors were **scattered across dates and corridors with no clear pattern**, the cleaned dataset retained **1,507 records**.

Trip duration (`tapOutTime` - `tapInTime`) and trip sequence (`stopEndSeq` - `stopStartSeq`) were also checked to ensure data consistency. All values were logical and positive, with no entry errors detected.

Data Exploration

1. How do bus stop usage patterns and traffic volumes affect route efficiency?

To explore this question, I used temporal and spatial analyses to understand demand distribution, congestion patterns, and route-level performance across Transjakarta services.

A. Time Series Pattern: Weekday vs. Weekend Line Plot

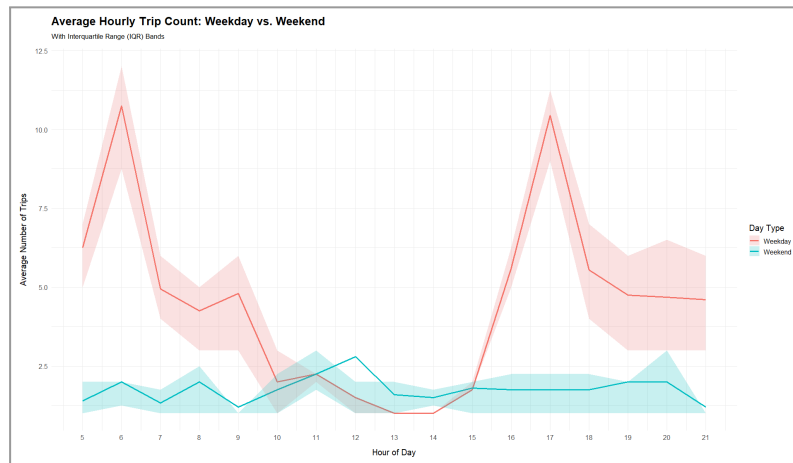


Figure 1. Line Chart of Average Hourly Trip Count: Weekday vs. Weekend

This plot illustrates distinct daily usage patterns. **Weekdays** show **pronounced peaks** around **5–6 AM** and **4–6 PM**, likely reflecting **work and school commutes**. In contrast, **weekends** display a flatter distribution, with a **modest increase around midday (12 PM)**, likely linked to **leisure activities**.

A **line plot** was used to highlight temporal trends, with **IQR bands** illustrating variability. **Weekday peaks** show narrow bands, indicating consistent demand, while wider bands in the evening reflect more varied travel behavior (e.g., late Friday trips). **Weekends** show generally wider bands, suggesting greater variability in user patterns. **Color coding** (red for weekdays, blue for weekends) helps distinguish usage trends, while larger, bold titles and thoughtful spacing improve readability and visual clarity.

The visual was created using `ggplot2` in R, with data processed using `mutate()`, `group_by()`, and `summarise()` in `dplyr`, with weekday/weekend classification handled by `lubridate`. Performed structured aggregation and filtering. Trips were grouped by hour and date, then aggregated by day type (weekday vs weekend). The entire process involved temporal feature extraction, group-wise summarisation, and visual encoding of variability

B. Heatmap of Demand: Hour vs. Day of Week

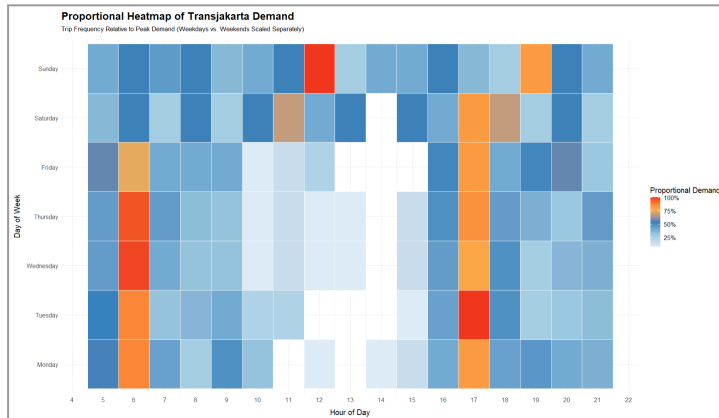


Figure 2. Proportional Heatmap of Transjakarta Demand

This heatmap **confirms weekday peak demand** around 6 AM and 5 PM, consistent with commuter routines shown in Figure 1. On the other hand **weekends reveal different behaviors**. Saturday peaks at 11 AM and 5–6 PM, while Sunday shifts later to 12 PM and 7 PM

To generate this heatmap, I grouped trip records by hour and day across individual dates, then aggregated the average hourly demand for each weekday and weekend group and to allow fair comparison, **values are scaled proportionally within weekday and weekend groups**, making distinct usage patterns easier to spot. This ensures each day is compared relative to its own peak, allowing clearer visibility of intra-group patterns without understating less frequent travel days. A **multi-hue color scale** was applied to enhance interpretability, allowing **clearer contrast** across demand levels and making peak zones **easier to identify**.

The visual was created using `ggplot2` in R, with `dplyr` and `lubridate` used to classify day types and normalize values proportionally across days using `mutate()` and `group_by()` before plotting.

C. Map-Based Demand Analysis

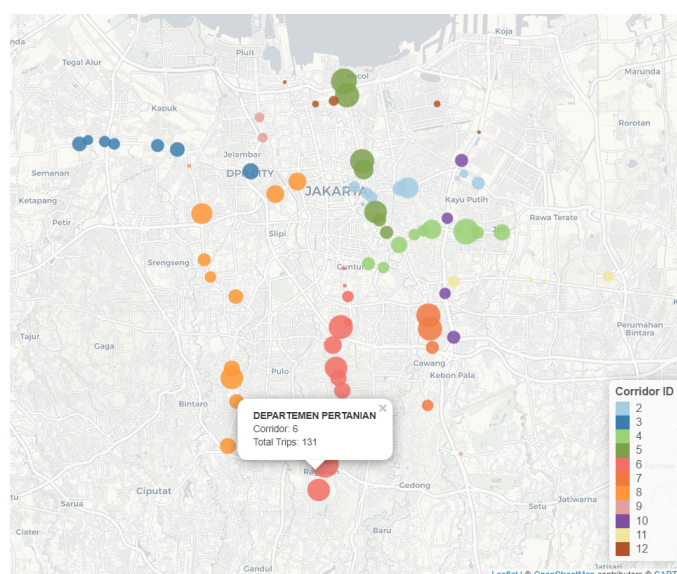


Figure 3. Map-Based Transjakarta Demand Analysis

This map reveals that certain corridors experience higher demand based on **combined tap-in and tap-out activity**. **Corridor 6** (Ragunan - Latuharhary) shows **consistently high usage** across all stops, followed by Corridors 8, 5, and 4. These findings highlight **potential priorities for bus stop upgrades**. Stops with significantly higher usage within a corridor may indicate key transfer points where improvements are more urgent.

Each corridor is assigned a **unique color for clarity**, while **circle size represents stop-level demand**, making it easy to compare usage across stops and corridors. A **grayscale map background helps keep the focus on demand patterns, avoiding visual clutter**. Popups were added with `paste0()` for quick insight on individual stops.

The map was built using `leaflet` in R, with `RColorBrewer` used to assign distinct color palettes per corridor. Coordinates were filtered to exclude missing data, and stop-level demand was scaled using a square root transformation to ensure proportional but visually readable circle sizes. This allowed for clearer differentiation between moderately and heavily trafficked stops.

D. Route Efficiency Index

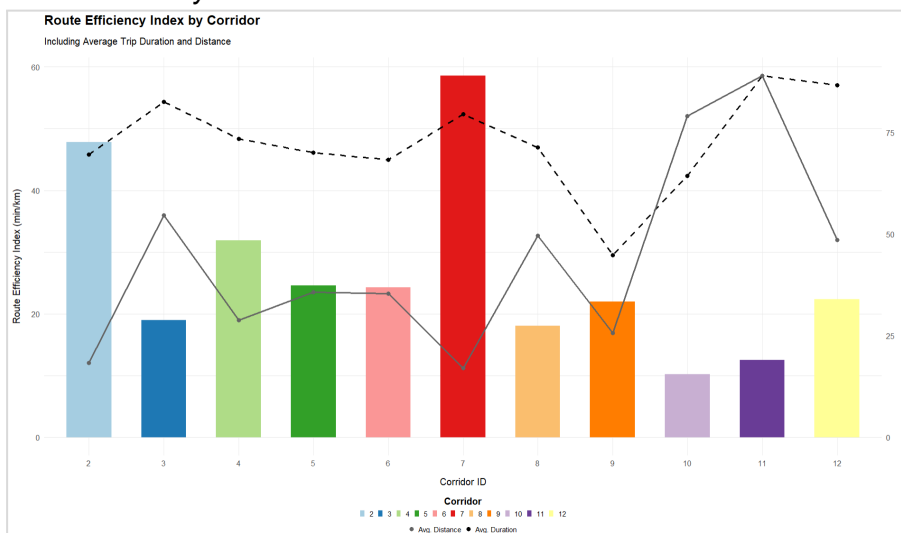


Figure 4. Mixed Chart of Route Efficiency Index by Corridor

This chart presents the Route Efficiency Index (REI), calculated as the average duration and distance per corridor. Corridors with lower REI values are more time-efficient for passengers (Wolfram, 2021).

The results indicate that Corridors 6 and 8 perform efficiently despite high demand (as shown in Figure 3), suggesting optimal routing and minimal delays. In contrast, Corridor 7 has the highest Route Efficiency Index (REI), yet also one of the longest average travel durations and the lowest stop density, pointing to inefficiency. Combined with the high-demand stops identified in Figure 3, this suggests that Corridor 7, particularly its busiest stops, may require infrastructure upgrades to better support passenger flow.

Including average duration and distance provides context beyond stop count or demand alone. Line styles (solid vs. dashed) differentiate metrics without visual clutter, while a bar chart supports corridor-level comparison. Color choices align with Figure 3 for visual consistency.

The visual was created using `ggplot2` in R, with `RColorBrewer` for the color palette and `geosphere` for distance calculations using the Haversine formula, `distHaversine()` function (Weisstein, n.d.), enabling an approximate real-world measurement of distance between bus stop coordinates. To enable visual comparison, both average duration and distance were rescaled to match the bar chart height and overlaid using dual-axis formatting, `geom_col()` and `geom_line()`.

2. What are the user profiles across routes, and how can this inform service improvements?

User behavior was explored based on demographics (age, sex), trip patterns, and payment methods.

A. Gender Proportion per Corridor

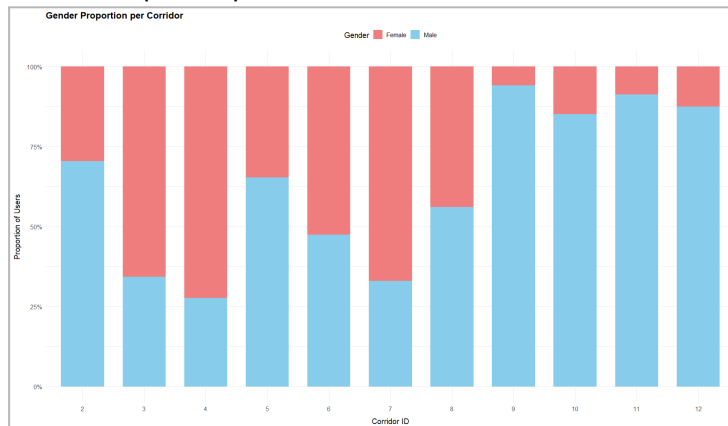


Figure 5. 100% Stacked Bar Chart of Gender Proportion per Corridor

This chart displays the gender proportion of passengers by corridor. The results show notable differences across the corridor such as, corridor 3, 4, and 7 dominated by female passengers, while corridor 9, 10, 11, and 12 are mostly used by male passengers. These findings can inform gender-focused service improvements, such as female-only sections, designated buses, or access to free sanitary products at key bus stops.

A 100% stacked bar chart was used to clearly display gender proportion without being influenced by passenger volume. The color choices follow common picturisation for male and female to support readability.

The chart was created using `ggplot2` in R, with `dplyr` and `scales` for gender categorisation, proportion calculation, and formatting. To prepare this chart, raw gender labels were first cleaned and standardized using conditional mapping (e.g., “M” mapped to “Male”) to ensure consistency across the dataset. The data was then grouped and aggregated by corridor, and each gender count was normalized to form proportions within each corridor.

B. Age Distribution by Corridor

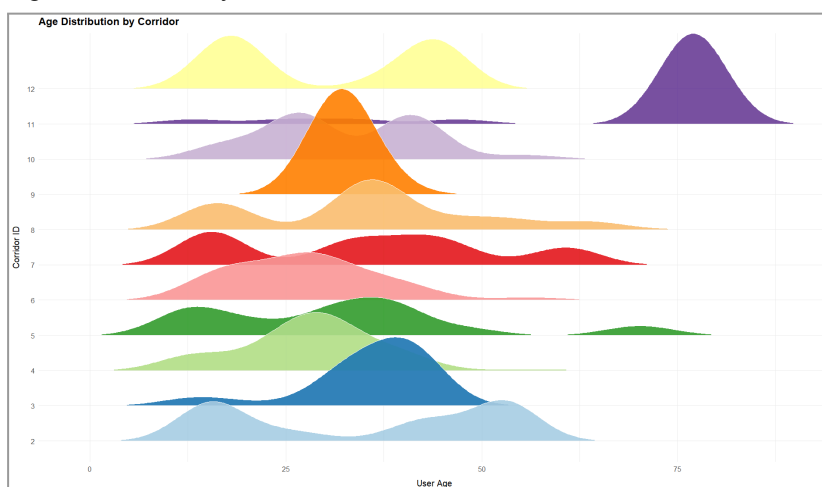


Figure 6. Ridge Density plot of Age Distribution by Corridor

This chart shows the age distribution of users across corridors. Most follow a similar pattern, with peak usage between ages 15 and 50, indicating a majority of commuters and students. Corridor 11 is skewed toward older users, suggesting a need for accessibility features like ramps and lifts. Corridors 2, 5, 7, and 12 show a higher proportion of users under 20, highlighting potential for student-focused improvements.

A ridge density plot was selected to show full distribution shapes for each corridor without overlapping, allowing clear comparisons in a single chart. The color palette is consistent with previous maps for corridor identification.

This visual was created in R using `ggplot2`, `ggridges`, `dplyr`, and `RColorBrewer` adapted from codeline and format from Kassambara (2020). The ridge density plot uses a method that smooths the age data into curves, showing the full distribution for each corridor. Unlike bar charts, this helps us see patterns like where the ages peak or spread out. Arranging the corridors vertically makes it easier to compare without overlap.

C. Payment Method by Corridor

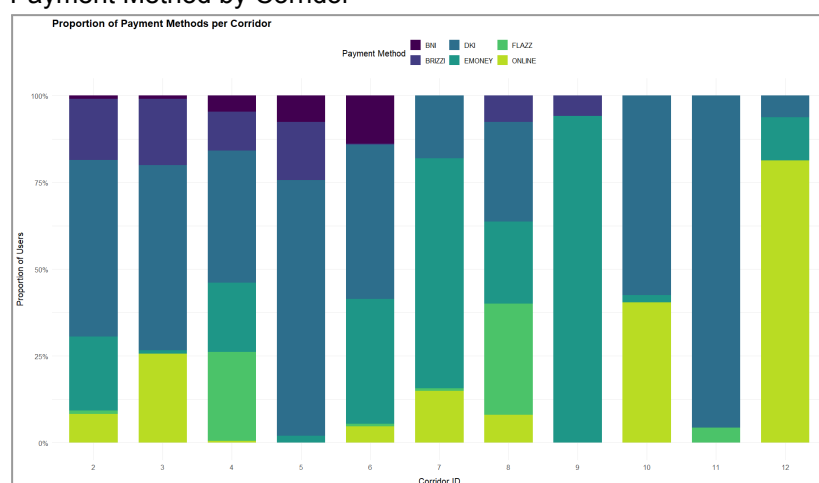


Figure 7. 100% Stacked Bar Chart of Payment Method Proportion by Corridor

This chart displays the proportion of payment methods used on each corridor. Results show distinct usage patterns: Corridors 12, 10, and 3 have high reliance on online payments, indicating a need to ensure seamless support for digital transactions. Corridors 9, 7, and 6 predominantly use e-money, while Corridors 11, 10, 5, 3, 2, and 6–8 show significant use of Bank DKI cards, highlighting key areas where tap-in machine compatibility should be prioritised.

A 100% stacked bar chart was chosen to emphasise proportional comparisons, independent of trip volume. The `viridis` palette was used for its color-blind friendliness and print clarity, ensuring accessibility and readability across audiences.

The chart was created in R using `ggplot2`, with `dplyr` and `scales` for data aggregation, transformation, and formatting. Grouping the data by corridor and payment method, then calculate the proportion.

3. Which bus stops require upgrades to accommodate increasing passenger demand?

To answer the core project question, which bus stops need improvement to accommodate increasing passenger demand, multiple insights from earlier visualizations were integrated through the following steps:

- 1) Exclude Figures 1 and 2, which focus on daily and hourly usage patterns, as they are more relevant for identifying optimal renovation timing rather than location.
- 2) Rank corridors by total trips (sum of tap-in and tap-out) based on Figure 3.
- 3) Rank individual bus stops by total trips from Figure 3.
- 4) Rank corridors by Route Efficiency Index (REI) from Figure 4 to identify inefficiencies.
- 5) Flag corridors with >50% female passengers based on Figure 5, to identify those that may benefit from gender-sensitive services.
- 6) Flag corridors with a median user age above 55 using Figure 6. The median is used instead of the mean due to the skewed age distribution, ensuring more representative results.
- 7) Identify the top 1–3 most-used payment methods per corridor using Figure 7, depending on whether each method exceeds a 25% usage threshold.
- 8) Join the resulting summary table with Dataset 2 to include bus stop age and infrastructure condition, enabling a more informed recommendation.

The final priority ranking is based on total trips per bus stop, with supporting insights from efficiency, demographics, and payment behavior to determine which top 5 bus stops should be prioritized for upgrades.

Table 1. Final Consideration Findings for Bus Stops Improvement

stopName	corridorID	trips	corridor_rank	bus_stop_age	female_service	older_access	payment_methods	rel_rank	notes
DEPARTEMEN PERTANIAN	6	131	1	17	Yes	No	DKI, EMONEY	7	PELICAN
ANCOL	5	110	2	17	No	No	DKI	8	PELICAN
SUNAN GIRI	4	107	4	17	Yes	No	DKI, FLAZZ	9	PELICAN
BIDARA CINA	7	104	5	17	Yes	No	EMONEY	11	JPO BAJA / EKSISTING (TANGGA SEMUA)
KUNINGAN TIMUR	6	104	1	17	Yes	No	DKI, EMONEY	7	JPO BETON / EKSISTING (TANGGA SEMUA)

Top 5 Bus Stops Needing Urgent Upgrades

Based on demand, infrastructure age, passengers demographics, and service performance, the following five stops are prioritized for improvement:

I. DEPARTEMEN PERTANIAN

- A. Old Infrastructure: Requires full structural review and potential reinforcement due to age.
- B. Female-Oriented Services: High proportion of female passengers suggests the need for gender-sensitive facilities.
- C. Payment Methods: Upgrade compatibility and Improve Bank DKI and E-Money tap response.
- D. REI Rank (7): Moderate inefficiency suggests the need to improve waiting comfort (e.g., more seating or shaded areas).
- E. Notes: Pelican crossing available – positive for pedestrian safety.

II. ANCOL

- A. Old Infrastructure: Requires full structural review and potential reinforcement due to age.
- B. Payment Methods: Upgrade compatibility and Improve Bank DKI tap response.
- C. REI Rank (8): Similar actions to Department Pertanian may improve commuter experience.
- D. Notes: Pelican crossing in place – accessibility adequate.

III. **SUNAN GIRI**

- A. Old Infrastructure: Requires full structural review and potential reinforcement due to age.
- B. Female-Oriented Services: High proportion of female passengers suggests the need for gender-sensitive facilities.
- C. Payment Methods: Upgrade compatibility and Improve Bank DKI and Flazz tap response.
- D. REI Rank (9) Indicates inefficient routing; optimize waiting areas and reduce bottlenecks.
- E. Notes: Pelican crossing supports safe access.

IV. **BIDARA CINA**

- A. Old Infrastructure: Requires complete structural and access review, including pedestrian access.
- B. Female-Oriented Services: High proportion of female passengers suggests the need for gender-sensitive facilities.
- C. Payment Methods: Upgrade compatibility and Improve E-Money tap response.
- D. REI Rank (11): One of the least efficient; waiting areas must accommodate delays.
- E. Notes: Pedestrian Overpass made from Steel / Existing not newly built (All Stairs) - Needs focus on improvement of access to bus stops.

V. **KUNINGAN TIMUR**

- A. Old Infrastructure: Requires complete structural and access review, including pedestrian access.
- B. Female-Oriented Services: High proportion of female passengers suggests the need for gender-sensitive facilities.
- C. Payment Methods: Upgrade compatibility and Improve Bank DKI and E-Money tap response.
- D. REI Rank (7): Moderate inefficiency suggests the need to improve waiting comfort (e.g., more seating or shaded areas)
- E. Notes: Pedestrian Overpass made from Steel / Existing not newly built (All Stairs) - Needs focus on improvement of access to bus stops.

Additionally, based on Figures 1 and 2, renovation work is best scheduled on weekdays between 10 AM and 3 PM, when passenger traffic is at its lowest.

This approach supports evidence-based infrastructure decisions, built using `left_join()`, `filter()`, `mutate()`, and `arrange()` from `dplyr`, and to finalize the analysis and prepare findings for presentation, the summary table of ranked bus stops was exported to Excel using the `writexl` package in R. This allowed for further formatting and annotation to clearly communicate the most urgent upgrade priorities.

Exporting to `.xlsx` also supports stakeholder-friendly outputs that can be easily shared or included in policy briefs and reports.

Conclusion

This project explored bus stop usage, passenger behavior, and infrastructure efficiency using Transjakarta transaction and location data. Each question is addressed below:

1. How do bus stop usage patterns and traffic volumes affect route efficiency?

While Figures 1 and 2 provide general insights into daily and weekly commuting patterns, Figures 3 and 4 directly address this question. The Map-Based Demand Analysis (Figure 3) and Route Efficiency Index (REI) by Corridor (Figure 4) reveal that uneven passenger distribution across stops affects overall route efficiency. For example, Corridors 7 and 2, which have the highest REI values, also show large disparities in demand between stops — which can create bottlenecks and reduce efficiency.

2. What are the user profiles across routes, and how can this inform service improvements?

Visualizations related to gender, age, and payment methods offer useful insights into user demographics and behavior. These patterns help identify where to prioritise upgrades — such as improved accessibility or gender-sensitive services.

3. Which bus stops require upgrades to accommodate increasing passenger demand?

A composite analysis synthesised route demand, infrastructure age, REI scores, and demographic needs, integrating insights from all prior visualizations. A final ranked list of five high-priority bus stops was defined. These findings provide actionable, data-informed guidance for improving infrastructure where it is most urgently needed.

Reflection

One key insight from the data wrangling and cleaning process is that even though the dataset is synthetic, it was modeled on real operational data and revealed the importance of handling missing values and noisy entries. A significant portion of the data had to be removed due to irrelevance or incompleteness, reinforcing the critical role of preprocessing in any data project.

Despite the dataset's limitations—such as its one-month scope, lack of detailed route trajectories, and missing stop interconnectivity information—the analysis still yielded meaningful, actionable insights. Including more comprehensive data in future studies would improve accuracy and allow for more robust conclusions, especially when conducting more complex analysis such as clustering or time series forecasting.

This project helped me appreciate that effective data visualization is not just about aesthetics, but about functionality and communication. Through this assignment, I explored different visualization techniques to deepen insights and better understand user patterns.

What I would improve next time:

- Spend more time understanding the data context and its real-world implications.
- Investigate how data is collected and what potential errors may occur.
- Plan the visualizations more strategically from the start to ensure each directly supports the research questions, rather than building insights step by step without a clear structure

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Acknowledgements

I used AI in the following ways:

(i) Refinement and Rephrasing

I used ChatGPT (<https://chatgpt.com/>) to refine sections of the Data Exploration Project (DEP) Part 2: Report to make it more concise and easier to read.

(ii) Idea Generation and Structural Support

I used ChatGPT (<https://chatgpt.com/>) to assist with any further improvement for the writing structure and deliverables.